Pablo Leyva

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Education

GEM Fellow
M.S. Applicant — Expected Fall 2027

New Jersey Institute of Technology
B.S. Applied Mathematics & Computer Science — Expected: May 2027

Experience

Research Engineer Intern, *Mechanistic Interpretibility Lab* — Sep 2026 – Present | Newark, NJ

- Advised by Kieran Murphy to build a **PyTorch** pipeline classifying all **144 GPT-2 attention heads** by cross-checking in **TransformerLens** and **Hugging Face** to quantify no-op attention behavior; 74% default to the BOS sink.
- Re-ran the 3,000 sequences of rare, common and byte-free tokens; 15 heads shifted oppositely in early and late layers.
- Running **causal experiments** of 1.2M sequences to isolate attention shift.

AI/ML Intern, *Apple* — May 2026 – Aug 2026 | Cupertino, CA

- Designed agentic-commerce use cases for Siri, using Apple Wallet and Event Passes data as context to anticipate user needs.
- Built a **multimodal synthesizer** fusing speech and **On-Screen Awareness** into instruction prompts.
- Proposed a screenless human-computer interface that delegates work to recursive subagents on the user's behalf.

Machine Learning Intern, *Angeles Investors* — Jan 2026 – May 2026 | New York City, NY

- Productized AI **agent swarm** powered by GPT-4o which led to a **$500K pre-seed deal** at a $48M pre-money valuation.
- Deployed an **asynchronous FastAPI** data ingestion pipeline via **Docker** that streamed structured results in real time, accelerating startup screening, due-dilligence and outreach.

President, *Society of Hispanic Professional Engineers (SHPE)* — Sep 2025 – May 2026 | Newark, NJ

- Led 13 officers fundraising **$40,000** across **90+** events year-round, growing membership to a record **250+**.
- Architected and shipped the chapter's **first iOS app** in **27 days** leading 3 engineers, scaling to **200+ users** in its **first month** with event registration, member details, and mentorship-program features.

AI/ML Intern, *Apple* — May 2025 – Aug 2025 | Cupertino, CA

- Deployed a **LangGraph** agent integrating **RAG** with **MCP** to reason over technical docs for Apple Pay Merchants.
- Used **Qdrant** for **multilingual similarity search** and **GraphQL** to fetch and manipulate data across internal systems.
- Built **LLM evaluation pipeline** benchmarking agent trajectories, tool-use success, groundedness, and hallucination rate.

Software Engineering Opeartions Intern, *Caterpillar Inc.* — May 2024 – Aug 2024 | Chicago, IL

- Implemented a complete **ETL workflow** in **Apache Spark** for Autonomous Vehicle Fleet Management, handling data validation, transformation, and storage in **PostgreSQL** with a **CI/CD** pipeline.
- Served end-to-end data pipelines via **FastAPI** microservices at **sub-100ms** latency for 10K+ daily requests.

Projects

Evolv | Devpost — *Flask, NumPy, Pandas, Python*

- Placed **1st of 546** teams at HackPrinceton, building an entire backend and predictive modeling system in under 12 hours.
- Built a **geospatial stochastic simulation** platform to forecast longterm regional impact.
- Productionized a **real-time Kubernetes pipeline** with LLM-generated reports, end-to-end running under 5 seconds.

Founder - HAI — *Transformers, PyTorch, pgvector, Redis, Docker, FastAPI, LangGraph, Python*

- Commercialized an AI-powered SAT prep platform, selling to NJ tutoring centers supporting **600 students** concurrently.
- Built **multimodal** input/output **pipelines** for PDFs, images, and text, combining **OCR**, vector search, semantic retrieval, prompt orchestration, and GPT-4o API to produce personalized questions, explanations, and study material.
- Implemented a **symbolic verifier** for mathemtical reasoning on CoT yielding **100%** solution correctness before delivery.

Criteo Uplift: Large-Scale Causal ML | GitHub | Kaggle — *LightGBM, EconML, scikit-learn, statsmodels, Python*

- Trained LR, RF, and **LightGBM** on **14M** ad-experiment users (0.29% conversions): **PR-AUC 0.226**, 78× base rate.
- **X-learner** uplift targeting found **60% more** incremental visits than conversion-probability targeting at a 5% budget.

Scholastic-LLM: On-Device Distillation | GitHub | Hugging Face — *MLX, PEFT, Hugging Face, Python*

- Distilled **Claude Sonnet 4.6** into an **8-bit Qwen 2.5 7B** via on-device **LoRA** in 12.7 GB, lifting task score **258%**.
- Built evals over 5 checkpoints; traced **DPO**'s zero gain to preference saturation; shipped best **SFT** via **PEFT**.

Skills

Languages: Python, SQL, C++, JavaScript/TypeScript, Swift
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LangGraph, RAG, MCP, Qdrant
Systems & Infrastructure: Docker, Git, Kubernetes, PostgreSQL, REST, GraphQL, FastAPI, Apache Spark, CI/CD, NumPy, Pandas